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EcoScan

REQUIREMENTS SPECIFICATION

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1. INTRODUCTION

1.1 Project Overview

EcoScan is a mobile-based application designed to promote sustainable consumption by helping users understand the environmental impact of everyday products. By scanning product barcodes, users can instantly access information about a product's ecological footprint and receive suggestions for more environmentally friendly alternatives.

The application also integrates motivational features such as challenges, rewards, and progress tracking to encourage users to adopt eco-friendly habits over time.

1.2 Objectives

The main objective of EcoScan is to raise awareness about the environmental impact of everyday consumer products and to support users in making more informed, sustainable purchasing decisions. The application aims to encourage eco-friendly behavior through the integration of motivational features such as challenges, rewards, and progress tracking. Additionally, EcoScan is designed to provide a simple, accessible, and user-friendly tool that can be easily integrated into users' daily routines, promoting long-term engagement with sustainable habits.

2. EXISTING SITUATION ANALYSIS

Currently, many users lack awareness regarding the environmental consequences of the products they purchase daily. Although several applications exist, they present notable limitations:

- **Yuka** focuses primarily on health aspects rather than environmental impact
- **Good On You** is limited to clothing products only
- **Too Good to Go** focuses on food waste reduction rather than product analysis

Additionally:

- These applications can be complex or not widely adopted
- There is no unified platform combining scanning, eco-analysis, and motivation
- Users often purchase products without realizing their environmental impact

3. PROBLEM IDENTIFICATION

The main problems identified are:

- Difficulty in identifying environmentally friendly products
- Lack of clear, accessible, and reliable information
- Low engagement in sustainable habits
- Absence of motivation systems such as rewards or challenges
- Limited tools that combine multiple eco-features in one application

4. PROPOSED SOLUTION

EcoScan addresses these issues through the following features:

- **Barcode Scanning:** Allows users to scan products instantly
- **Eco Impact Score:** Displays environmental impact indicators (e.g., carbon footprint, recyclability)
- **Sustainable Alternatives:** Suggests eco-friendly substitutes for scanned products
- **User Profile:** Tracks user activity, achievements, and progress
- **Community Challenges:** Encourages users to complete eco-tasks and earn rewards

This solution aims to combine simplicity, usability, and engagement into one platform.

5. FUNCTIONAL REQUIREMENTS

The system must provide the following functionalities:

- Allow users to create and manage accounts
- Enable barcode scanning of products
- Display of product details and environmental impact
- Suggest eco-friendly alternatives for scanned products
- Allow users to participate in challenges
- Track user progress (points, completed challenges, history)
- Provide feedback and rewards (points, badges)
- Allow administrators to manage products, challenges, and system data

6. NON-FUNCTIONAL REQUIREMENTS

The system must satisfy the following constraints:

- **Performance:** Fast response time for barcode scanning and data retrieval
- **Usability:** Simple, intuitive, and user-friendly interface
- **Security:** Protection of user data and secure authentication
- **Compatibility:** Mobile compatibility (Android/iOS)
- **Availability:** System should always be accessible
- **Scalability:** Ability to handle increasing users and product data

7. SYSTEM ACTORS

- **User (EcoUser):**
    - Scans products
    - Views eco impact and alternatives
    - Participates in challenges
    - Tracks progress
- **Administrator:**
    - Manages products and alternatives
    - Creates and updates challenges
    - Maintains system data

8. USE CASE OVERVIEW

Main use cases include:

- Scan Product
- View Product Information
- View Eco Impact Score
- Get Alternative Products
- Participating in Challenge
- Track Progress

These use cases describe interactions between users and the system and are represented in the UML diagrams.

9. DATA MODEL OVERVIEW

The system is based on the following main entities:

- **EcoUser:** Stores user information
- **Product:** Contains product details and eco impact
- **Alternative:** Represents eco-friendly substitutes
- **Challenge:** Defines tasks users can complete
- **Progress:** Tracks user achievements
- **EcoUser_Challenge:** Links users to challenges

Relationships:

- One product can have multiple alternatives
- One user can participate in multiple challenges
- Each user has associated progress data

10. CONCLUSION

EcoScan provides an innovative and practical solution to promote environmentally responsible consumption. By combining product scanning, eco-analysis, and user engagement features, the system encourages users to adopt sustainable habits in their daily lives.

The proposed system is simple, scalable, and impactful, making it a valuable tool for raising environmental awareness and driving behavioral changes.